

OPUS DEVIA

Your path. Proven by you.

Comprehensive Technical Documentation

Version 1.0 - June 2026

Built with React 19 + TypeScript + Vite + Supabase + Tailwind CSS 4

1. Overview

Opus Devia is a gamified business-mentorship web application that guides users from self-discovery through a structured entrepreneurial journey. It combines AI-driven psychological profiling, personalised roadmap generation, XP-based gamification, and a dual-mode AI mentor chat to help users build a business aligned with their natural strengths and personality archetype.

The application is designed as a mobile-first Progressive Web App (PWA) styled with a dark, military-grade luxury aesthetic featuring glassmorphism, crimson accents, and custom animation systems. It runs on a React 19 front-end with Supabase handling authentication, database, and serverless Edge Functions.

2. What the App Does

2.1 User Journey

1. SIGN UP / SIGN IN - Users authenticate via email and password through Supabase Auth.
2. ONBOARDING QUESTIONNAIRE - Users answer 6 deep psychological and practical questions about their background, decision-making style, energy, discipline, ambitions, and blockers.
3. AI PROFILE GENERATION - The Roadmap Generator edge function sends answers to Gemini 2.5 Flash (or OpenRouter) which returns a comprehensive business-psychology profile:
 - Archetype classification (Closer, Creator, Strategist, Operator, Maverick)
 - Extracted scores (skill alignment, infrastructure fit, risk motivation, etc.)
 - Path compatibility ratings across 10 business paths
 - Psychology profile (loves, hates, triggers, disruptors, habits, beliefs)
4. ARCHETYPE REVEAL - A cinematic animation reveals the user's archetype with a description.
5. PERSONALISED ROADMAP - A 3-phase, 15-task roadmap is generated from templates matched to the user's archetype and recommended business path.
6. HOME DASHBOARD - The daily hub shows level, XP, streak, roadmap progress, and the current priority task.
7. MENTOR CHAT - An AI mentor (or assistant) provides real-time guidance costing XP per message.
8. XP ECONOMY - Users earn XP by completing roadmap tasks; XP is spent on mentor interactions.

2.2 Core Features

- AI Archetype Profiling via Gemini 2.5 Flash / OpenRouter
- Personalised 3-phase business roadmap with 15 actionable tasks
- XP-based gamification system (earn, spend, balance tracking)
- Dual-mode AI Mentor Chat (Mentor: strategic guidance, Assistant: quick help)
- Daily streak tracking with weekly visual indicator
- Level system based on cumulative earned XP
- Roadmap progress bar with milestone tracking
- Priority task display on home dashboard
- Draft saving during onboarding (resume later)
- Phase auto-unlock when all tasks in a phase complete (85% threshold)
- Tier-gated features (Free, Builder, Operator, Founder)
- Cinematic animations and transitions throughout the UI

3. How It Works

3.1 Authentication Flow

Authentication is handled by Supabase Auth using email/password. The `useAuth()` hook manages session state, user data, and onboarding completion status. The router (React Router DOM v7) protects routes based on authentication and onboarding state:

- Unauthenticated users are redirected to `/signin`
- Authenticated users without completed onboarding go to `/onboarding`
- Authenticated users with onboarding complete go to `/home`
- Mentor chat (`/mentor`) is accessible from the home screen

3.2 Onboarding Pipeline

The onboarding module has three stages managed as a state machine:

STAGE 1 - QUESTIONNAIRE: Six questions presented one at a time with slide animations. Each requires minimum 20 characters. Users can save drafts to Supabase (`draft_sessions` table) and resume later. Progress bar tracks completion.

STAGE 2 - PROCESSING: A pulsing logo animation with rotating status messages while the roadmap-generator Edge Function calls Gemini AI. The function:

- a) Checks user tier and XP preflight for paid tiers
- b) Sends all 6 answers to Gemini with a detailed system prompt
- c) Receives a structured JSON profile with archetype scores, path ratings, and psychology data
- d) Generates phase templates matched to archetype + path
- e) Writes roadmap, phases, tasks, and persistent memory to Supabase

STAGE 3 - REVEAL: Staggered cinematic entrance animation revealing the archetype name and description. User clicks Enter to complete onboarding and navigate to Home.

3.3 XP & Gamification

The XP system is the core game mechanic:

- **EARNED XP:** Awarded for completing roadmap tasks (varies by difficulty: 15-100 XP)
- **PURCHASED XP:** Available via tier upgrades
- **ROLLOVER XP:** Carried from previous periods
- **RESERVED XP:** Committed but not yet consumed
- **AVAILABLE XP** = earned + purchased + rollover - reserved

$LEVEL = \text{floor}(\text{earnedXP} / 100)$, minimum level 1.

MENTOR COSTS: Mentor mode costs 5 XP/message, Assistant mode costs 2 XP/message. XP is deducted optimistically before the API call and refunded on error.

TIERS: Free (one roadmap, capped tokens), Builder (2000 tokens), Operator (3000 tokens), Founder (4000 tokens). Paid tiers use the billing-manager Edge Function for preflight XP checks.

3.4 Mentor Chat

The mentor chat is a dual-model AI conversation system:

- **MENTOR MODE:** Strategic business guidance, roadmaps, decisions, deep thinking

- ASSISTANT MODE: Quick help, task assistance, lighter interactions

Messages invoke the 'model-router' Edge Function which routes to the appropriate AI model based on the active mode. Each conversation is tracked as a 'session' in the sessions table (created on mount, closed on unmount or model switch). XP is deducted per message. Insufficient XP triggers a TierGateModal prompting upgrade.

Switching between Mentor and Assistant modes closes the current session, clears the chat window with a fade transition, and opens a new session.

3.5 Roadmap System

Roadmaps consist of 3 phases, each containing 5 tasks (15 total). Phases follow a Foundation -> Validation -> Scaling pattern. When all tasks in a phase are marked complete, the next phase auto-unlocks (if unlock_condition is 'auto'). A separate 85% early-unlock check allows mentor override for phase unlocking.

The roadmap generator uses a template library for known archetype+path combinations (e.g., Closer+Freelance, Creator+Freelance, Strategist+Consulting) with detailed, actionable task descriptions. Unknown combinations use a fallback generator that creates phases from AI-suggested major tasks.

3.6 Database Schema (Key Tables)

users	- id, email, display_name, tier, onboarding_complete, free_roadmap_used
user_xp	- user_id, earned, purchased, rollover, reserved_xp
user_streaks	- user_id, streak_type, current_count, last_activity_date
onboarding_sessions	- user_id, responses, status, computed_profile, current_archetype
draft_sessions	- user_id, responses, current_question, updated_at
roadmaps	- user_id, title, archetype, status, current_phase, roadmap_data
roadmap_phases	- roadmap_id, phase_number, title, status, unlock_condition
tasks	- roadmap_id, roadmap_phase_id, title, is_completed, xp_reward
sessions	- user_id, session_type (mentor/assistant), status
memory_persistent	- user_id, strengths, weaknesses, execution_patterns, roadmap_state

4. Architecture & Structure

4.1 Tech Stack

- Frontend: React 19.2, TypeScript 6.0, Vite 8.0
- Routing: React Router DOM v7.17
- Styling: Tailwind CSS 4.3 + custom CSS with glassmorphism system
- Backend / DB: Supabase (Auth, PostgreSQL, Edge Functions)
- AI: Gemini 2.5 Flash via native API or OpenRouter proxy
- Mobile: Capacitor 8.4 (configured, ready for native builds)
- Package Manager: npm

4.2 Project Directory Structure

```
opus-devia/
  index.html           # Entry HTML
  package.json         # Dependencies & scripts
  vite.config.ts       # Vite configuration
  tailwind.config.ts   # Tailwind theme (colors, tokens)
  tsconfig.json        # TypeScript config
  .env.example         # Env var reference (VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY)
  src/
    main.tsx           # App entry, renders Router + animated background
    App.tsx            # Legacy placeholder (unused)
    index.css          # Global styles, design system, animations
    router/
      index.tsx        # BrowserRouter + route protection logic
    lib/
      supabase.ts      # Supabase client init (env vars, proxy fallback)
    hooks/
      useAuth.ts       # Auth state, session, onboarding status
      useMentorChat.ts # Chat state machine, XP deduction, model-router calls
    pages/
      SignIn.tsx       # Email/password sign-in form
      SignUp.tsx       # Email/password registration form
      Onboarding.tsx   # 3-stage onboarding (questionnaire/processing/reveal)
      Home.tsx         # Dashboard (level, streak, roadmap, priority task)
      MentorChat.tsx   # Dual-mode AI chat interface
    components/
      home/
        Header.tsx     # Top bar with username + XP
        LevelHero.tsx  # Level badge ring + archetype display
        StreakPill.tsx # Daily streak counter + weekly day circles
        RoadmapBar.tsx # Completion % + milestone tracker + progress bar
        PriorityTask.tsx # Current priority task card with XP reward
        TaskSectionHeader.tsx # Section header for task area
        MentorButton.tsx # CTA card linking to mentor chat
        BottomNav.tsx  # Fixed bottom navigation (Home/Quest/Roadmap/Profile)
      onboarding/
        QuestionCard.tsx # Single question with textarea + slide animation
        ProcessingScreen.tsx # Rotating status messages + pulsing logo
        ArchetypeReveal.tsx # Cinematic archetype reveal with entrance animation
      mentor/
        ChatHeader.tsx # Model switcher + XP balance display
        ChatWindow.tsx # Message list with user/AI bubbles
        MessageInput.tsx # Text input + send button
    ui/
```

```

    XPCounter.tsx      # XP amount display component
    TierGateModal.tsx  # Upgrade prompt modal
  assets/
    logo.png           # App logo (KING branding)
    hero.png           # Hero image
    badges/            # Level badge SVGs (lv_1_10.svg, etc.)

```

4.3 Supabase Edge Functions

```

supabase/functions/
  roadmap-generator/index.ts # Main onboarding + roadmap generation function
  Actions:
    - generate: Full roadmap generation from questionnaire answers
    - check_early_unlock: 85% completion check for phase unlock
    - auto_unlock_phase: Cascade phase unlock after task completion
  Features:
    - Gemini 2.5 Flash integration (native + OpenRouter)
    - Template library for archetype+path combinations
    - Fallback phase generator for unknown combinations
    - Tier gating with billing-manager preflight
    - Token caps per tier (2000-4000)
    - Persistent memory updates after generation

  model-router/index.ts      # Routes chat messages to mentor/assistant AI
  billing-manager/index.ts    # XP preflight/reservation for paid features

```

4.4 Design System

The application uses a cohesive dark design system:

COLORS:

- Background: #000000 (pure black)
- Primary accent: #9a0000 (crimson) with #DC143C (crimson-light) for hover states
- Cards: #181920 with glassmorphism effects
- Text: #F5F5F5 (white-soft), #A8A8A8 (silver-1), #C0C0C0 (silver-2)
- Progress: #1d9e75 (green for success states)

GLASS EFFECTS:

- backdrop-filter: blur(16px) saturate(120%)
- Linear gradient base with inset highlight/shadow
- Specular top highlight, thick bottom shadow
- Noise texture overlay (2px pattern at 1.5% opacity)

EASING CURVES:

- --ease-snappy: cubic-bezier(0.16, 1, 0.3, 1) [entrance, hover-in]
- --ease-return: cubic-bezier(0.25, 0.1, 0.25, 1) [hover-out]
- --ease-breath: cubic-bezier(0.4, 0, 0.2, 1) [ambient animations]

ANIMATIONS:

- spotlightBreath (14s) - Hero spotlight pulsing
- badgeFloat (8s) - Badge ring gentle float
- assistPulse (6s) - Assist orb breathing
- haloGlow (6s) - Red ambient halo
- progressGlow (4s) - Progress bar glow
- activeDock (4s) - Active nav item glow
- currentDayPulse (3s) - Current streak day pulse

- fadeUp (0.8s)
- logo-pulse (2s)
- Page entry staggered animation
- Logo breathing during processing

5. Route Map

Route	Auth Required	Onboarding Required	Component
/signin	No	-	SignIn
/signup	No	-	SignUp
/onboarding	Yes	Not completed	Onboarding
/home	Yes	Completed	Home
/mentor	Yes	Completed	MentorChat
* (catch-all)	-	-	Redirect to /signin

6. Data Flow Diagrams

6.1 Onboarding & Roadmap Generation

```
User -> Onboarding.tsx
  |-> QuestionCard (6 questions, draft saving to draft_sessions)
  |-> ProcessingScreen
  |-> supabase.functions.invoke('roadmap-generator', { action: 'generate', userId, answers })
      |-> Check tier + XP preflight (billing-manager)
      |-> callGeminiForProfile(answers, maxTokens)
      |-> generatePhaseTemplates(archetype, path, majorTasksHint)
      |-> INSERT roadmap, phases (3), tasks (15)
      |-> UPDATE onboarding_sessions (status: ready)
      |-> UPSERT memory_persistent
      |-> Return { archetype, success }
  |-> ArchetypeReveal (cinematic animation)
  |-> UPDATE users.onboarding_complete = true
  |-> navigate('/home')
```

6.2 Mentor Chat Flow

```
User -> MentorChat.tsx
  |-> useMentorChat(userId)
      |-> createSession(model) -> sessions table
      |-> fetchXP() -> user_xp table
      |-> sendMessage(content)
          |-> XP preflight (check balance >= cost)
          |-> Optimistic XP deduction
          |-> supabase.functions.invoke('model-router', { action, session_id, message })
          |-> On success: display AI reply, confirm XP deduction
          |-> On error: refund XP, show error on message
      |-> setActiveModel(mode): close old session, create new session
  |-> ChatHeader (model switcher, XP display)
  |-> ChatWindow (message list)
  |-> MessageInput (send, XP cost label)
  |-> TierGateModal (shown on insufficient XP)
```

6.3 Home Dashboard Data

```
Home.tsx -> fetchHomeData()
  |-> users: display_name
  |-> user_xp: earned, purchased, rollover, reserved_xp
  |-> user_streaks: current_count, last_activity_date -> derive weekly days
  |-> onboarding_responses: archetype value
  |-> roadmaps (active) -> roadmap_phases (incomplete, ordered) -> tasks (incomplete, first)
  |-> tasks: total count + completed count -> roadmapCompletion %
  |-> Assemble -> setData() -> render children (LevelHero, StreakPill, etc.)
```


7. Security & Environment

7.1 Secret Management

All API keys and service credentials are managed through environment variables:

- VITE_SUPABASE_URL / VITE_SUPABASE_ANON_KEY: Client-side, prefixed with VITE_ for exposure
- SUPABASE_URL / SUPABASE_SERVICE_ROLE_KEY: Server-side only (Edge Functions)
- GEMINI_API_KEY / GEMINI_BASE_URL / GEMINI_MODEL: AI provider credentials
- Additional provider keys follow *_API_KEY, *_BASE_URL, *_MODEL naming conventions

Secrets are never hardcoded. Development secrets live in .env.local (gitignored). Production secrets are set via Supabase dashboard or push-secrets-from-env.ps1 script.

7.2 Supabase Client Safety

The Supabase client (src/lib/supabase.ts) implements a proxy fallback: if environment variables are missing or invalid, the client returns a Proxy object that throws on any method call, preventing import-time crashes while allowing the React tree to render. The router handles auth state correctly in this degraded state (user appears logged out).

7.3 Row-Level Security

All Supabase tables use Row-Level Security (RLS) policies to ensure users can only access their own data. The Supabase service role key is used exclusively in Edge Functions and never exposed to the client. Unique partial indexes enforce constraints like one active roadmap per user.

8. Scripts & Development Tooling

```
npm run dev      # Start Vite dev server with HMR
npm run build    # TypeScript check + Vite production build
npm run lint     # ESLint across project
npm run preview  # Preview production build locally

scripts/
  push-secrets-from-env.ps1      # Push .env.local secrets to Supabase Edge Functions
  set-edge-secrets.example.ps1  # Example flow for setting Edge Function secrets
  apply-migration.ps1          # Apply Supabase database migrations
  install-supabase-cli.ps1     # Install Supabase CLI
  push-secrets.mjs              # Node.js secret pushing script
```

9. Key File Reference

Frontend (24+ source files)

- src/main.tsx**
Application entry point. Renders animated background + Router.
- src/router/index.tsx**
React Router config with auth/onboarding route guards.
- src/lib/supabase.ts**
Supabase client factory with env validation and proxy fallback.
- src/hooks/useAuth.ts**
Auth state hook: session, user, loading, onboardingComplete.
- src/hooks/useMentorChat.ts**
Chat logic: sessions, XP deduction, model-router invocation.
- src/pages/SignIn.tsx**
Sign-in form with email/password, Supabase auth, navigation.

src/pages/SignUp.tsx
Registration form with password confirmation.

src/pages/Onboarding.tsx
3-stage onboarding state machine (questionnaire/processing/reveal).

src/pages/Home.tsx
Dashboard aggregating 6 Supabase queries into a unified view.

src/pages/MentorChat.tsx
Dual-model chat container with fade transitions and tier gating.

src/index.css
1573-line design system: variables, glass effects, 12+ keyframe animations.

src/components/home/LevelHero.tsx
Level badge ring, XP display, archetype label.

src/components/home/StreakPill.tsx
Daily streak flame counter + weekly day circles.

src/components/home/RoadmapBar.tsx
Completion percentage + milestone tracker + progress bar.

src/components/home/PriorityTask.tsx
Priority task card with XP reward and time display.

src/components/home/MentorButton.tsx
Navigation CTA to mentor chat.

src/components/home/BottomNav.tsx
Fixed bottom navigation bar (4 tabs).

src/components/onboarding/QuestionCard.tsx
Single question with textarea, slide animation, draft save.

src/components/onboarding/ProcessingScreen.tsx
Rotating status messages during AI generation.

src/components/onboarding/ArchetypeReveal.tsx
Cinematic staggered-reveal animation for archetype.

src/components/ui/XPCounter.tsx
Reusable XP amount display component.

tailwind.config.ts
Custom color palette (crimson, silver, white-soft, panel).

Backend (Edge Functions)

supabase/functions/roadmap-generator/index.ts
1467 lines. Main generation engine: Gemini AI call, template matching, phase/task insertion, memory updates, tier checks, early-unlock logic.

supabase/functions/model-router/index.ts
Routes mentor/assistant chat messages to appropriate AI models.

supabase/functions/billing-manager/index.ts
XP preflight validation and reservation for paid features.

10. Summary

Opus Devia is a sophisticated gamified business mentorship platform that leverages AI to create personalised entrepreneurial roadmaps. It combines psychological profiling, structured task management, XP-based gamification, and conversational AI mentorship into a cohesive mobile-first web application.

The architecture follows modern best practices:

- Frontend: React 19 with TypeScript, Vite, and Tailwind CSS 4
- Backend: Supabase (Auth, PostgreSQL, Edge Functions)
- AI: Gemini 2.5 Flash with OpenRouter as an alternative provider
- Security: RLS policies, environment-based secrets, proxy fallbacks
- Design: Custom glassmorphism system with 12+ keyframe animations

Total codebase: ~24 frontend source files + 3 Edge Functions + database migrations + helper scripts, all organised for maintainability and scalability.